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Section: BSCS 3B

Subject: CSST102 – Basic Machine Learning

Assessment: Assessment Task No. 1 - Introduction to Machine Learning

From Data to Prediction: Understanding Machine Learning-Based Social Media Recommendation

Chosen ML Application:

Social Media Recommendation - a system that suggests people or accounts that a user may want to follow based on their interests, activities, connections, and previous interactions.

Problem Description:

Social media platforms have many users and accounts, which can make it difficult for users to find people or accounts that they may be interested in following. To solve this problem, a social media recommendation system can use the user's previous activities, interests, connections, and interactions. From these data, the system can identify patterns and recommend people or accounts that the user might want to follow. This can make it easier for users to discover relevant accounts without having to search through many profiles manually.

Input and Output Identification:

The table below identifies the application, problem, inputs, outputs, patterns, and data needed.

Question	Your Answer
1. What is the application?	Social media recommendation. For example, a system that suggests people or accounts that a user may want to follow.
2. What problem are you trying to solve?	Recommend relevant people or accounts to a user based on their interests, connections, and previous activities.
3. What are the inputs?	Number of mutual friends, shared interests, followed pages or groups, previous interactions, and user activities.
4. What is the expected output?	A list of recommended people or accounts that the user may be interested in following.
5. What patterns or relationships might the ML system learn?	Users with more mutual friends and shared interests may be more likely to follow or connect with each other.

6. What data would the system need?	User profiles, friendship/following information, mutual friend data, user interests, groups and pages followed, previous interactions, and historical follow/connection data.
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Traditional Programming vs. Machine Learning:

In traditional programming, the programmer needs to know and write the rules that the computer will follow. The flow of traditional programming is **Input** → **Algorithm/Rules** → **Output**. For a social media recommendation system, a programmer could create rules such as, "**If two users have at least 5 mutual friends, recommend the person.**" Another rule could be, "**If two users have at least 3 shared interests, recommend the person.**" The programmer would need to create many rules for different situations.

For example, if a user has **6 mutual friends and 4 shared interests**, the programmed rules could recommend that account.

In Machine Learning, the flow is different. It uses **Input + Output/Data** → **Machine Learning Algorithm** → **Learned Relationship**. Instead of manually creating every rule, the algorithm is given many examples of previous user behavior. It studies the inputs and outputs and learns the relationship between them. After training, the model can use new information to make a prediction.

This makes Machine Learning useful because social media behavior can be complicated and difficult to describe using manually written rules.

Mini Dataset:

Below is a simple dataset showing the relationship between the number of mutual friends, shared interests, and whether the user followed the recommended account.

Observation	Mutual Friends	Shared Interests	Followed?
1	8	5	Yes
2	6	4	Yes
3	2	1	No
4	3	2	No
5	7	4	Yes
6	1	1	No
7	5	3	Yes

8	4	1	No
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Pattern/Relationship:

The pattern that I notice in the dataset is that users with more mutual friends and shared interests are generally more likely to follow the recommended account.

For example, users with 5 or more mutual friends and 3 or more shared interests mostly resulted in "Yes." On the other hand, users with fewer mutual friends and shared interests mostly resulted in "No."

For a new example:

Mutual Friends	Shared Interests	Followed?
7	5	Yes

The predicted output is Yes because the user has many mutual friends and shared interests, which is similar to the examples that resulted in "Yes."

For this simple dataset, we can represent the relationship using a basic scoring function:

Recommendation Score = Number of Mutual Friends + Number of Shared Interests

For example:

7 + 5 = 12. So the recommendation score is **12**.

A higher score in this simplified example suggests that the user may be more likely to follow the recommended account. This is only a simplified mathematical representation used to demonstrate the relationship between the inputs and output in the mini-dataset. A real social media recommendation system would learn more complex relationships from many different types of user data.

Explanation of the Training Process:

How does a Machine Learning algorithm learn from data?

The algorithm first receives input data such as the number of mutual friends and shared interests, together with the correct output, which is whether the user followed the recommended account. During training, the algorithm studies many examples and compares its predictions with the actual results. It adjusts its model so that its predictions become closer to the correct outputs and gradually

learns the relationship between the inputs and outputs. After training, the model can use new data that it has not seen before to make a prediction.

The process is:

Input → Output → Training → Learning the Relationship → Prediction

For this example:

Mutual Friends + Shared Interests → Followed/Not Followed → Training → Learn User Behavior → Predict Follow/No Follow

Short Reflection:

Why is learning from data useful instead of manually programming every possible rule?

Learning from data is useful because real-world situations can be too complicated to describe using individual rules. In social media, users have different interests, behaviors, friends, and interactions, so creating a rule for every possible situation would be difficult and time-consuming. Machine Learning allows the computer to discover patterns from many examples and use those patterns to make predictions. Therefore, learning from data is more practical for recommendation systems than manually programming every possible rule.